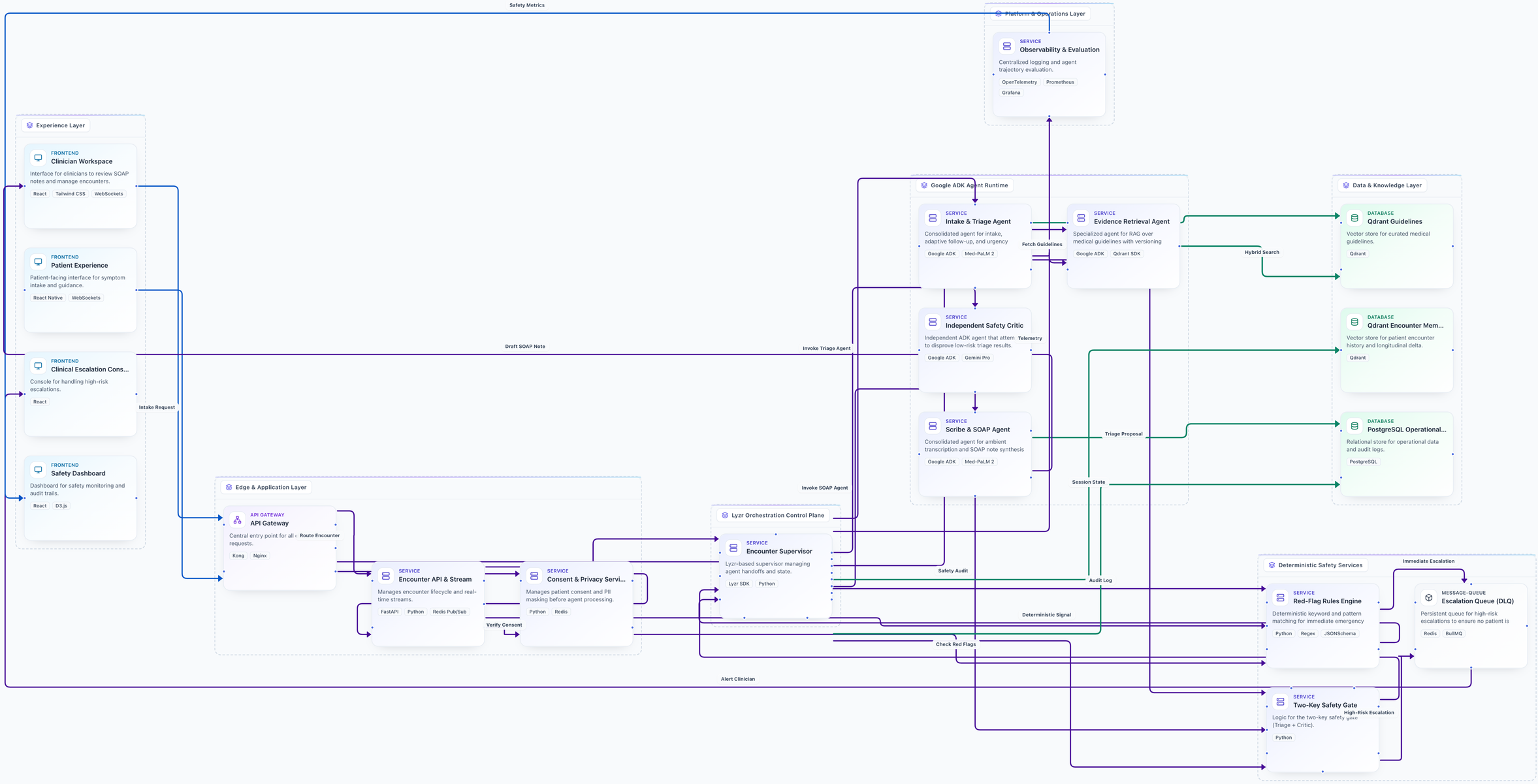
Safety Metrics
Platform & Operations Layer
SERVICE
Observability & Evaluation
Centralized logging and agent trajectory evaluation.
OpenTelemetry
Prometheus
Grafana
Experience Layer
FRONTEND
Clinician Workspace
Interface for clinicians to review SOAP notes and manage encounters.
React
Tailwind CSS
WebSockets
FRONTEND
Patient Experience
Patient-facing interface for symptom intake and guidance.
React Native
WebSockets
FRONTEND
Clinical Escalation Cons...
Console for handling high-risk escalations.
React
FRONTEND
Safety Dashboard
Dashboard for safety monitoring and audit trails.
React
D3.js
Draft SOAP Note
Intake Request
Google ADK Agent Runtime
SERVICE
Intake & Triage Agent
Consolidated agent for intake, adaptive follow-up, and urgency
Google ADK
Med-PaLM 2
SERVICE
Evidence Retrieval Agent
Specialized agent for RAG over medical guidelines with versioning
Google ADK
Qdrant SDK
Fetch Guidelines
SERVICE
Independent Safety Critic
Independent ADK agent that attem
Google ADK
Gemini Pro
Telemetry
SERVICE
Scribe & SOAP Agent
Consolidated agent for ambient transcription and SOAP note synthesis
Google ADK
Med-PaLM 2
Triage Proposal
Session State
Invoke Triage Agent
Invoke SOAP Agent
Data & Knowledge Layer
DATABASE
Qdrant Guidelines
Vector store for curated medical guidelines.
Qdrant
Hybrid Search
DATABASE
Qdrant Encounter Mem...
Vector store for patient encounter history and longitudinal delta.
Qdrant
DATABASE
PostgreSQL Operational...
Relational store for operational data and audit logs.
PostgreSQL
Edge & Application Layer
API GATEWAY
API Gateway
Central entry point for all
Route Encounter
requests.
Kong
Nginx
SERVICE
Encounter API & Stream
Manages encounter lifecycle and real-time streams.
FastAPI
Python
Redis Pub/Sub
SERVICE
Consent & Privacy Servi...
Manages patient consent and PII masking before agent processing.
Python
Redis
Verify Consent
Lyzr Orchestration Control Plane
SERVICE
Encounter Supervisor
Lyzr-based supervisor managing agent handoffs and state.
Lyzr SDK
Python
Safety Audit
Audit Log
Deterministic Signal
Check Red Flags
Deterministic Safety Services
Immediate Escalation
SERVICE
Red-Flag Rules Engine
Deterministic keyword and pattern matching for immediate emergency
Python
Regex
JSONSchema
MESSAGE-QUEUE
Escalation Queue (DLQ)
Persistent queue for high-risk escalations to ensure no patient is
Redis
BullMQ
SERVICE
Two-Key Safety Gate
Logic for the two-key safety gate (Triage + Critic).
Python
High-Risk Escalation
Alert Clinician